

Quarter-Tone Keyboard (QTKB) – Build Guide

This is the active build guide for the synth / semi-weighted workstream and the single active technical source for geometry, prototype rules, and build decisions.

Public shorthand for GitHub-facing material: QTKB.

The previously documented piano-like intermediate branch is no longer active and is preserved under docs/archive/SteinwayLike/.

Active Branch Context

- Active branch: June 2026 onward
- Primary feel reference: Access Virus C
- Keybed family reference: Fatar-style synth action
- Active mechanical category: synthesizer / semi-weighted with integrated aftertouch

Active design anchors:

- shallower package than the archived piano-like branch
- consistent feel across all 122 keys
- quarter-tone keys may differ in force and geometry, but must remain inside the same overall action family
- aftertouch is part of the base key-travel design, not an add-on

Active Reference Interpretation

Current externally supported reference facts:

- the keyboard Virus C variant (Virus KC) is a 61-key instrument
- it is described publicly as 61 semi-weighted keys
- it supports velocity and monophonic aftertouch at the keyboard level
- current Fatar synth families such as TP/9S, TP/8S, and TP/8SK are explicitly listed as synthesizer keyboards and are offered with 61-key variants and aftertouch options

Current working conclusion:

- the player-facing MT dimensions can remain unchanged
- the player-facing QT top-side dimensions can also remain unchanged for now
- the redesign focus is the internal mechanics, packaging depth, return system, sensor placement, and aftertouch integration

Current interpretation of the reference aftertouch category:

- public Fatar information suggests pressure-oriented aftertouch sensing with very little intentional extra travel
- for this project, Hall-based aftertouch should therefore use a small but deliberate reserve travel instead of blindly copying a near-zero-travel pressure-strip concept
- the active design question is not a published Fatar aftertouch travel number, but the reserve travel that is musically useful in this custom Hall-based action

Frozen Player-Side Dimensions

These player-facing dimensions remain active until contradicted by prototype evidence:

- MT octave width: 165.0 mm
- MT white key width at front: 23.6 mm
- MT white visible length: 150.0 mm
- MT black key width: 12.5 mm
- MT black visible length: 95.0 mm
- MT black key height above white: 12.0 mm
- QT width: ≤ 12.5 mm
- QWT top-side span: $Y = 86-170$ mm
- QWT rest height above MWT plane: +12 mm
- QST top-side span: $Y = 116-170$ mm
- QST rest height above MWT plane: +24 mm
- QT front-travel target band: about 8-10 mm

Build Phase Overview

Phase	Content	Status
1	Define synth-action geometry baseline	Active
2	Build single-key synth-action test bench	Open
3	Verify Hall sensor and aftertouch package	Open
4	Derive quarter-tone integration for the synth branch	Open
5	Build one-octave mock-up	Open
6	Electronics prototype	Open
7	Firmware integration	Open
8	Full mechanical iteration	Open

Phase 1: Active Geometry Baseline

Coordinate system rule:

- white-key front top edge at rest: $0 = (0, 0)$
- Y points rearward from the player
- Z points upward relative to the white-key top reference plane

Active geometry rules:

- use main-key pivot baseline $Y = 210$ mm, $Z = -12$ mm

- QT keys are short secondary levers on the shared main pivot axis, not on a separate QT pivot axis
- do not silently revive archived defaults such as $Y = 210 \text{ mm}$, $Z = -24 \text{ mm}$ or the earlier two-axis proposal $MT = (195, -12)$, $QT = (235, 6)$
- treat splint, buffer, and sensor as one compact reaction group per key family while keeping the four family lanes MWT, MST, QWT, and QST separate from each other
- keep aftertouch inside the core travel design from the beginning

Active support and stop architecture:

- use pins for lateral guidance and anti-twist control
- use a simple laid-in pivot axle with spacer washers as the current key-spacing baseline
- allow the axle and spacer stack to be bonded into a removable holder cradle with UHU Endfest
- use replaceable Tough Resin bearing inserts rather than making the structural frame the wear part
- keep only the rear zero stop behind the pivot; the terminal end stop stays on the split-pin / guide line
- preserve the tail volume behind the pivot for zero-stop packaging, spring hardware, and optional trim mass

Active packaging rules:

- 6 mm maximum lever body within a 6.8 mm per-key lane
- 16-18 mm first-pass box height over most of the hidden lever body
- 20-22 mm local build-up around pivot, spring eye, guide-bushing pocket, and optional weight pocket
- prototype the lever, cradle, stop, and support geometry in FDM first before introducing metal construction

Current key-family naming:

- MWT = Main Whole-tone key
- MST = Main Semitone key
- QWT = Quarter Whole-tone key
- QST = Quarter Semitone key

Phase 2: Single-Key Test Bench

Immediate prototype goal:

- one main key in the new synth-action category
- one matching quarter-tone key concept for packaging comparison
- Hall sensor position and magnet geometry representative of the active branch
- integrated aftertouch travel reserve as part of the base design

The first prototype must answer:

- whether the intended synth-like feel is reachable with the chosen lever and return concept
- how much travel reserve is available for aftertouch without destroying the main playing feel
- how much depth the active branch actually needs

Current first-pass key-family tables:

MWT Key

Field	Current value	Note
Motion concept	Main lever	Standard white-key family
Front reference point	$Y = 0 \text{ mm}, Z = 0 \text{ mm}$	White-key front top edge at rest
Pivot axis	$Y = 210 \text{ mm}, Z = -12 \text{ mm}$	Active working baseline
Lever width	6 mm maximum	Shared current lever width inside a 6.8 mm per-key lane
Box height over hidden lever	16-18 mm	FDM baseline
Box height at pivot section	20-22 mm	Local reinforced height
Normal front travel	9.8 mm	Current white-key reference under the fixed full-aftertouch geometry
Key angle at normal travel	2.67 deg	From $\text{asin}(9.8 / 210)$
Key angle at full aftertouch	3.2 deg	Fixed user value for the full-aftertouch end position
Spring start	$20 \times 4 \times 0.5 \text{ mm}, 22.0 \text{ mm}$ eye-to-eye at rest	First fixed start value without trim weights
Zero stop	$Y = 285 \text{ mm}$	Hard printed rail
End stop	Split-pin / guide line	Terminal stop line

MST Key

Field	Current value	Note
Motion concept	Main lever	Standard black-key family
Front reference point	$Y = 53.5 \text{ mm}, Z = 12.0 \text{ mm}$	Black-key top reference plane at rest
Pivot axis	$Y = 210 \text{ mm}, Z = -12 \text{ mm}$	Shared main-key pivot
Lever width	6 mm maximum	Shared current lever width inside a 6.8 mm per-key lane
Box height over hidden lever	16-18 mm	FDM baseline
Box height at pivot section	20-22 mm	Local reinforced height
Normal front travel	10.0 mm	Current standard black-key travel
Key angle at normal travel	3.66 deg	From $\text{asin}(10.0 / 156.5)$
Key angle at full aftertouch	4.3 deg	Fixed user value for the full-aftertouch end position

Field	Current value	Note
Spring start	20 x 4 x 0.4 mm, 21.5 mm eye-to-eye at rest	First fixed start value without trim weights
Zero stop	Y = 285 mm	Hard printed rail
End stop	Split-pin / guide line	Terminal stop line

QWT Key

Field	Current value	Note
Motion concept	Short secondary lever	Shared main pivot axis, no separate QT pivot axis
Front reference point	Y = 86 mm, Z = 12 mm	QWT top front edge at rest
Pivot axis	Y = 210 mm, Z = -12 mm	Shared main-key pivot
Lever arm to front reference	124 mm	210 - 86
Top-side width	<= 12.5 mm	Current QT width limit
Visible length	84 mm	Top-side span Y = 86-170 mm
Rest height above MWT plane	+12 mm	Whole-step quarter-tone row
Normal front travel target	About 8-10 mm	Current target band
Key angle at normal travel	3.70-4.63 deg	From $\arcsin(8/124)$ to $\arcsin(10/124)$
Key angle at full aftertouch	5.6 deg	Fixed user value for the full-aftertouch end position
Box height at pivot section	20-22 mm	Shared first-pass reinforcement zone; use the upper end where the guide-bushing pocket sits in the 6 mm body
Local QT body height	16-18 mm	Same first-pass box height as the main-key family; increase only where splint, spring, or local insert packaging demands it
Spring start	25 x 4 x 0.4 mm, 26.0 mm eye-to-eye at rest	First fixed start value without trim weights

QST Key

Field	Current value	Note
Motion concept	Short secondary lever	Shared main pivot axis, no separate QT pivot axis
Front reference point	Y = 116 mm, Z = 24 mm	QST top front edge at rest
Pivot axis	Y = 210 mm, Z = -12 mm	Shared main-key pivot

Field	Current value	Note
Lever arm to front reference	94 mm	210 - 116
Top-side width	≤ 12.5 mm	Current QT width limit
Visible length	54 mm	Top-side span $\gamma = 116-170$ mm
Rest height above MWT plane	+24 mm	Semitone quarter-tone row
Normal front travel target	About 8-10 mm	Current target band
Key angle at normal travel	4.88-6.11 deg	From $\text{asin}(8/94)$ to $\text{asin}(10/94)$
Key angle at full aftertouch	7.2 deg	Fixed user value for the full-aftertouch end position
Box height at pivot section	20-22 mm	Shared first-pass reinforcement zone; use the upper end where the guide-bushing pocket sits in the 6 mm body
Local QT body height	16-18 mm	Same first-pass box height as the main-key family; increase only where splint, spring, or local insert packaging demands it
Spring start	25 x 4 x 0.4 mm, 26.5 mm eye-to-eye at rest	First fixed start value without trim weights

Shared Summary

Field	Current value	Note
Return-spring line	$\gamma = 235$ mm	Common first spring line for all four key types
Zero-stop material	Printed hard rail, preferred ASA, PETG acceptable for fast tests	Height should be defined by the hard rail even when a damping layer is added
Zero-stop contact width in γ	2.5-3.0 mm	Narrow, defined contact band for repeatable rest height
Zero-stop contact width across lever	4-5 mm on a 6 mm lever	Keeps contact defined without using the full width as a hard impact plane
Zero-stop damping option	Thin constrained solid TPU or PORON strip/profile in front of the hard rail	Preferred first damping approach; do not use a hollow profile as the default reference-defining stop

Field	Current value	Note
Zero-stop height adjustment	Shim or tape on hard rail	Prototype-friendly replacement for paper punching / key punching Can stay in the original active zone; shrink-wrapped weight may also damp the zero stop when weights are used, but should not define the stop geometry
Optional trim-mass zone	Y = 255-285 mm	Current available test weight; usable on a 6 mm lever if mounted lengthwise in Y and upright in Z
First trim-mass test piece	5 g, 19 x 11 x 3 mm	Fits the restored active trim-mass zone well; use a local carrier or reinforced lever section if needed
First trim-mass test placement	Center about Y = 275 mm	Standing lengthwise packaging is acceptable because height is available behind the pivot
Weight packaging note	19 mm in Y, 3 mm across lever width, 11 mm in Z	Separate reaction-group lanes by key family; within each lane splint, buffer, and sensor are now packaged together
Functional package lanes	MWT, MST, QWT, QST	Current placement reference for each splint / buffer / sensor group
Reaction-group reference	5 mm behind the key front edge in the full-aftertouch tilt state	Current brass-splint packaging distance inside the reaction group
Reaction-group sensor / splint spacing	22 mm centre-to-centre	Keeps the PTFE-lined bore closer to horizontal inside the key while sharing angular error between pin and bushing
Guide-splint angle rule	Splint axis and splint-bushing axis each use half of the maximum local key tilt angle	Minimize corrective post-bending; use bending only for small final fitting corrections
Guide splint orientation	Aligned with the local key travel	Current active guide-pin baseline; small corrective bending remains possible if needed during fitting
Guide splint material and diameter	Brass, 2.5 mm	Do not force one common splint length for MWT, MST,
Guide splint length	Variable by key family	

Field	Current value	Note
Guide bushing block format	4 x 6 x 32 mm	QWT, and QST; set the effective length per key type First-pass rectangular outer package
PTFE-lined length inside bushing	18 mm for MT, 15 mm for QT	Current fixed guide-liner baseline PTFE liner should sit in a form-locking pocket; roughen the outer PTFE surface first and use cyanoacrylate only as position-locking support, not as the primary structural element
PTFE retention concept	Pocket plus adhesive	
Guide bushing overlenth	+2 mm relative to splint	Gives end margin beyond the effective splint length for each key family
Guide-bushing governing case	QWT and QST	MT keys still have more surrounding material at the splint location; the narrow QT keys are the actual strength and packaging limit case
QT bushing-side wall condition	About 1 mm per side at 6 mm key width	Acceptable for the first prototype if the bushing zone is locally reinforced in Z and all pocket transitions keep generous radii
Full-aftertouch reference angles	MWT 3.2 deg, MST 4.3 deg, QWT 5.6 deg, QST 7.2 deg	Fixed user values for the geometric layout at the maximum aftertouch stop
Guide half-angle values	MWT 1.6 deg, MST 2.15 deg, QWT 2.8 deg, QST 3.6 deg	Splint axis and splint-bushing axis each use half of the local full-aftertouch angle
Guide-angle data status	Positions still open per key family	Angular values are now fixed; exact locations still need to be entered separately for MWT, MST, QWT, and QST
Aftertouch buffer element	3 mm EPDM solid round cord, about 60 Shore A	Current first candidate for the terminal end-stop aftertouch buffer in both families
Buffer geometry split	QT: integrated into the lever geometry; MT: mostly housed in the raster	MT buffers should protrude only about 4 mm so they do

Field	Current value	Note
		not strike too wide into the buffer
Buffer groove	Shallow round groove, start range $R = 4.5-5.0$ mm, $t = 1.5$ mm	Applies to the integrated QT lever-side buffer geometry; MT keeps the buffer mostly in the raster
Buffer retainer / contact layer	2 mm felt strip	Common contact layer for end stop and aftertouch onset
Felt state at normal key pressure	About 1 mm compressed thickness	Defines the end of the normal playing stroke
Additional aftertouch reserve	About 1-1.5 mm local cord compression	Remaining path comes from EPDM compression after the felt-defined end point
Geometry reference state	Full-aftertouch end position	Current layout work is now referenced to the maximum aftertouch stop rather than to the normal-travel endpoint
Front aftertouch reserve interpretation	Just under 2 mm target at the key front for each key family	MWT is now aligned via about 11.8 mm full front travel and about 9.8 mm normal front travel
Key magnet size	Cylindrical 4 x 3 mm, grade N45	Current active Hall-sensing magnet on each key
Hall sensor to ferromagnetic hardware clearance	30 mm preferred, 25 mm minimum, centre-to-centre	Applies to steel screws, washers, frame parts, and other ferromagnetic hardware in all directions, including below the sensor; brass splints may be closer
Prototype material strategy	FDM first	No metal construction required in the first iteration

What the first prototype should verify specifically:

- whether the rear zero stop gives a stable and repeatable rest height without introducing bounce
- whether the hard zero-stop rail at $Y = 285$ mm is easy enough to adjust in practice with tape/shim build-up
- whether a thin constrained solid TPU or PORON damping layer reduces noise without adding noticeable rest-height drift or hysteresis
- whether the fixed first spring set at $Y = 235$ mm gives a useful initial force spread across MWT, MST, QWT, and QST
- whether shrink-wrapped trim weights in the 255-285 mm zone provide sufficient acoustic damping at the zero stop when weights are used

- whether the split-pin line is the correct place for the final end stop in the simplified synth branch
- whether the 3 mm EPDM cord plus 2 mm felt stack produces the desired compliance and noise behavior
- whether 2.5 mm brass guide splints with key-family-specific lengths plus 18 mm MT PTFE lining and 15 mm QT PTFE lining avoid tip entry and edge rubbing at maximum tilt for all four key families
- how far the four package lanes actually need to diverge in Y and Z
- whether a spring line around $Y = 230-240$ mm gives the right return feel without too much force increase over the stroke
- whether an optional trim-mass pocket around $Y = 255-285$ mm is sufficient if balancing mass is needed later
- whether a single 5 g trim weight at about $Y = 275$ mm produces a useful feel change without over-biasing the low-force action

Preserved Technical Decisions

The following remain active independent of the mechanical branch change:

- Kassel quarter-tone layout
- one Hall sensor per key
- Teensy-based real-time scanning front-end
- USB-MIDI and 5-pin DIN in phase 1
- elastomer-based aftertouch reserve rather than FSR force sensing

Mechanical Tasks Now Considered Closed in the Active Root

The following topics are no longer maintained in this active build guide because they belong to the archived branch:

- piano-like main-key pivot geometry around the archived long-lever proportion
- archived QT pivot package derived from the deeper action family
- MP5-based proportional reference as an active anchor

These remain available only in docs/archive/SteinwayLike/.

Phase 1 Open Data And Prototype Questions

Data still worth collecting:

- exact Access Virus C keyboard family or the closest reliable Fatar equivalent
- representative front travel of the reference action
- key length and effective pivot distance of the reference action
- aftertouch onset position and useful reserve travel
- force impression over the main stroke and aftertouch zone
- packaging depth required for the reference category

Immediate CAD / prototype questions:

- What is the minimum viable main-key depth for the active branch?
- Can the Hall sensor sit in front of, above, or behind the pivot without creating feel penalties?
- How should QT keys be packaged so they remain consistent with the shallower main-key action?
- Which return-force concept best matches a Fatar-like synth feel in a custom build?

Immediate Deliverables

- identify usable Access Virus C / Fatar reference measurements
- define first active main-key travel target
- define first active aftertouch travel target
- choose provisional pivot and sensor packaging concept for the shallow action
- derive first QT layout sketch relative to the new main-key geometry
- define first lane map for MWT, MST, QWT, and QST for split pins, buffers, and sensors
- define the first lever cross-section around a 6 mm maximum body width and 16-18 mm box height
- prototype the rear zero-stop, spring line, and optional trim-mass zone in FDM before any metal work

Documentation Rule

Any new active geometry decision should be documented in this build guide before it is propagated into docs/CAD/ or prototype steps.